

MA374 Financial Engineering Lab

Lab 08 Report

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1 Objective

To compute European call and put option prices using the Black-Scholes-Merton model and study their behavior with respect to time, underlying asset price, and model parameters.

2 Black-Scholes Formula

$$C(t, s) = sN(d_1) - Ke^{-r(T-t)}N(d_2)$$

$$P(t, s) = Ke^{-r(T-t)}N(-d_2) - sN(-d_1)$$

$$d_1 = \frac{\ln(s/K) + (r + \frac{\sigma^2}{2})(T-t)}{\sigma\sqrt{T-t}}, \quad d_2 = d_1 - \sigma\sqrt{T-t}$$

3 Parameters Used

- $T = 1$
- $K = 1$
- $r = 0.05$
- $\sigma = 0.6$

4 2D Plots

4.1 Call Option Price vs s

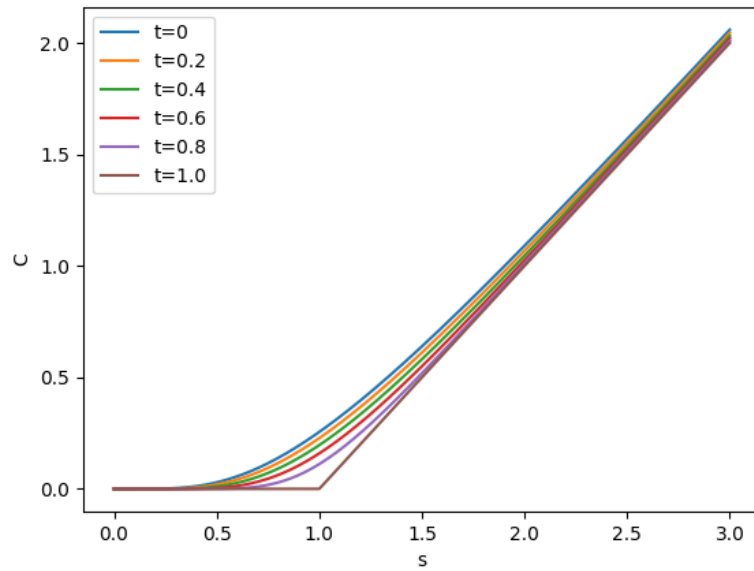


Figure 1: Call option price as a function of asset price s for different time values.

The graph shows that call prices increase with s . As time approaches maturity, the curves converge towards intrinsic value ($s - K$).

4.2 Put Option Price vs s

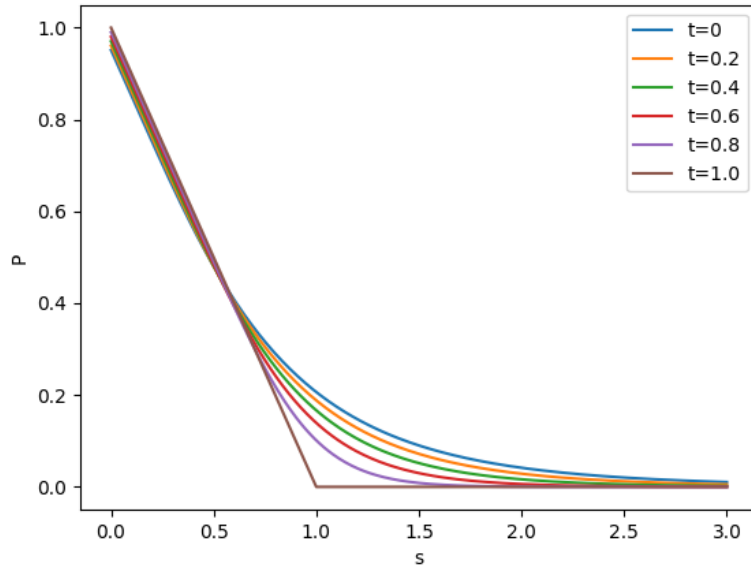


Figure 2: Put option price as a function of asset price s for different time values.

The put price decreases as s increases. For large s , the put value approaches zero, reflecting reduced likelihood of payoff.

5 3D Surface Plots

5.1 Call Option Surface

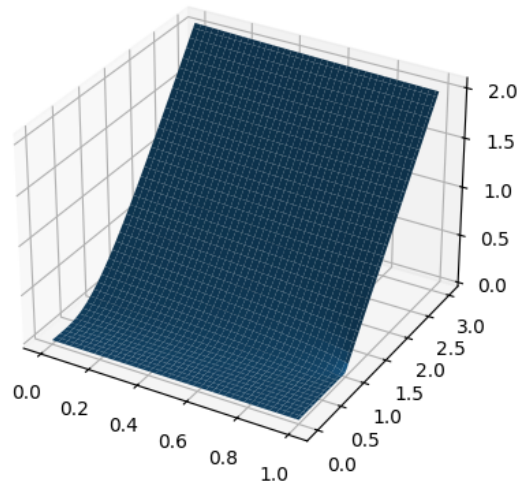


Figure 3: 3D surface of call option price as a function of time t and asset price s .

The surface shows convex growth in s and smoother variation over time. Near maturity, the surface aligns with intrinsic payoff.

5.2 Put Option Surface

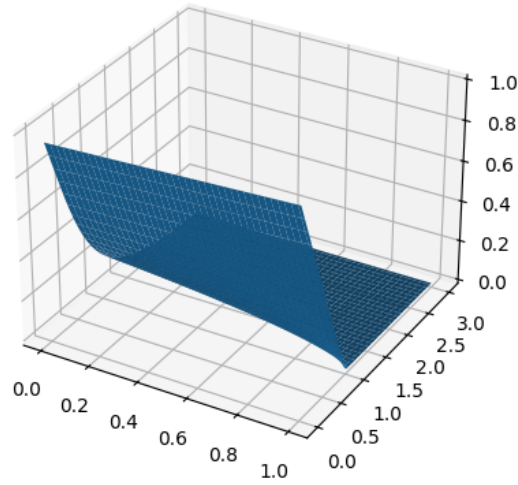


Figure 4: 3D surface of put option price as a function of time t and asset price s .

The put surface decreases with increasing s and shows stronger curvature near maturity due to reduced time value.

6 Sensitivity Analysis

6.1 Effect of Volatility σ

σ	Call Price	Put Price
0.1	0.0680	0.0193
0.2	0.1045	0.0557
0.4	0.1802	0.1315
0.6	0.2552	0.2065
0.8	0.3282	0.2794
1.0	0.3984	0.3496

6.2 Effect of Interest Rate r

r	Call Price	Put Price
-0.02	0.2282	0.2484
0.00	0.2358	0.2358
0.02	0.2435	0.2237
0.05	0.2552	0.2065
0.08	0.2672	0.1903
0.10	0.2752	0.1800

6.3 Effect of Strike Price K

K	Call Price	Put Price
0.8	0.3475	0.1085
0.9	0.2979	0.1540
1.0	0.2552	0.2065
1.1	0.2188	0.2651
1.2	0.1877	0.3292

7 Observations

- As volatility σ increases, both call and put prices increase due to higher uncertainty.
- As interest rate r increases, call prices increase while put prices decrease.
- As strike price K increases, call prices decrease and put prices increase.
- Call price increases with s , while put price decreases.
- For large s , call price becomes approximately linear and put price approaches zero.
- As $t \rightarrow T$, option prices converge to intrinsic values.
- Option price curves are convex, indicating asymmetric payoff structure.
- Time impact is strongest near $s \approx K$ where uncertainty is highest.
- Volatility has the most significant effect among all parameters.

8 Conclusion

The Black-Scholes model effectively explains the dependence of option prices on underlying parameters. Volatility plays the most dominant role in pricing. While the model provides clear insights through analytical and graphical methods, its assumptions may limit real-world applicability.